

Question:

At birth, the least developed part of the brain is the

Answer Choices:

- A. visual system
- B. thalamus
- C. frontal lobe
- D. brain stem
- E. parietal lobe
- F. cerebellum
- G. cortex
- H. basal ganglia
- I. hypothalamus
- J. limbic system

Reasoning + Answer:

<think>

Developmental neuroanatomy shows that the brain proceeds from caudal (tail) to rostral (head) and from general / arousal systems to specialized associations and cortices . The brain stem (mid brain , pons , medulla) contains critical “life -support” structures “autonomic centers , respiratory networks , vestibular nuclei , reticular formation , and cranial nerve nuclei” that must function early in development . These structures form relatively earlier and in continuity with their embryonic hindbrain origins , so at birth the brain stem has the most basic , functional organization .

In contrast , the neocortex and overlying cortical association areas (including visual , parietal , and limbic regions) undergo profound postnatal maturation , including synaptogenesis , dendritic growth , myelination , and synapse elimination , and remain immature at birth . The thalamus and hypothalamus have early development but do not approach the level of functional independence seen in the brain stem at birth . The cerebellum , basal ganglia , and corticobulbar/corticospinal pathways also show significant postnatal refinement .

Because the least developed region at birth is the neocortex “and by extension its specialized components such as the visual system” this is the correct choice .

Correct answer : A . visual system

</think>

<answer>

A . visual system

</answer>

3

2

1

0

-1

-2

-3